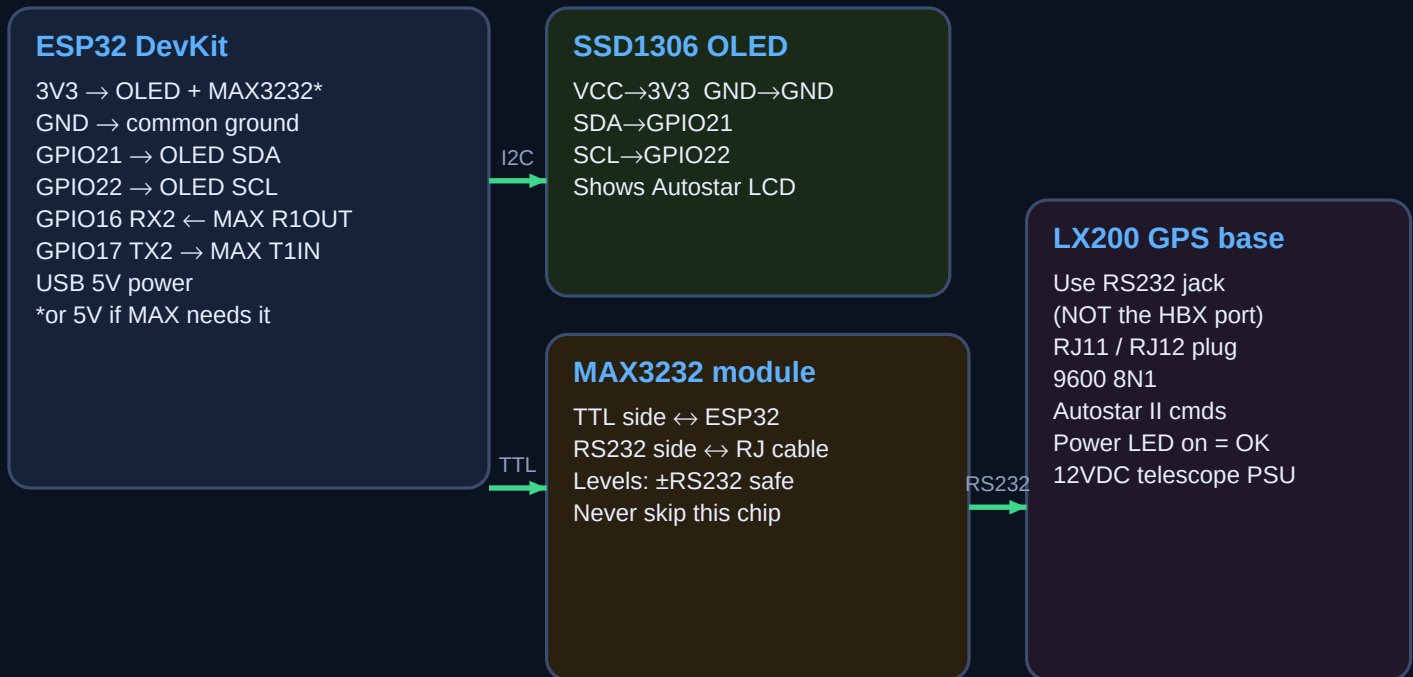


Meade LX200 GPS — ESP32 wiring

Product: LX200 GPS fork mount · Autostar II protocol · RS232 RJ jacks on base panel



Meade #507-style RS232 cable (RJ11 ↔ MAX3232)

Looking into LX200 GPS RS232 jack (clip/tab down, pins 1..6 left→right):

RJ pin	Signal on mount	Wire to MAX3232 RS232 side
1,2,6	unused	leave open
3	RX into telescope	← MAX3232 T1OUT (RS232 TX from ESP)
4	GND	↔ MAX3232 GND
5	TX from telescope	→ MAX3232 R1IN (RS232 RX to ESP)

TTL side of MAX3232: T1IN←ESP32 GPIO17 R1OUT→ESP32 GPIO16 VCC/GND
Handbox Autostar II stays on HBX if you want — ESP32 uses RS232 only.

CRITICAL: Plug into the jack labeled RS232 — do NOT use HBX for this ESP32 link.

HBX is the coiled Autostar II handbox bus. RS232 is the computer / LX200 command port.

If timeouts: swap RJ pins 3↔5 once, confirm 9600 baud, mount powered (red LED on).

Compatible product: Meade LX200 GPS (Autostar II / Smart Mount Technology base panel).

Bring-up checklist — LX200 GPS

After wiring MAX3232 + RJ into RS232

Power & ports on the fork base (from your photos)

OFF/ON rocker · 12VDC · RETICLE · HBX · RS232 · RS232 · AUTOGUIDER ...

1. Power mount 12V — red LED on control panel.
2. Insert RJ into either RS232 jack (both speak LX200 serial).
3. Leave HBX free OR plug physical Autostar II there (optional).
4. Flash TelescopeController.ino · Serial 115200 shows AP IP.
5. Phone → Wi-Fi ESP32_Telescope_AP / telescopepassword
6. Browser <http://192.168.4.1> — Autostar II virtual handbox
7. Tap Get RA / Get Dec — reply prints on green LCD + raw box

Autostar II keys → LX200 commands

▲▼■ = :Mn# :Ms# :Mw# :Me# (hold) · release sends :Qn# etc.
GO TO = set RA/Dec of library object + :MS# (+ tracking :Te#)
SPEED = cycle :RG# :RC# :RM# :RS# (Guide/Center/Find/Slew)
Get RA / Get Dec = :GR# / :GD# Ver = :GV# STOP = :Q#
SS / STAR / M / NGC = filter object library on the webpage
ENTER = load target · MODE = cycle RA/Dec · Target · Alt/Az · Link